Victor WOELFFEL

Final-year engineering student at Télécom Paris, currently on exchange at Polytechnique Montréal. I am interested in embedded systems and in understanding how artificial intelligence algorithms work.

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Education

Télécom Paris — Institut Polytechnique de Paris *Sep. 2023 — present*
Master of Science in Engineering, Embedded Systems track *Paris-Saclay, France*

- Graduation expected in December 2027. Exchange at Polytechnique Montréal, August 2026–January 2027.
- Embedded systems: C, Rust, STM32, processor architecture and hardware verification.
- Distributed systems: Java, Akka, MQTT, client-server architectures, HTTP and socket communication.
- Artificial intelligence: supervised learning, PyTorch and reinforcement learning.

Professional Experience

Emerton Data *Mar. — Aug. 2026*
Data Science Consultant Intern *Paris, France*

- Optimization: simplified the formulation and adapted the constraints of a complex system using Gurobi. These changes reduced computation time by 30%. Only my contribution is described here, as the project is confidential.
- AI agents: developed agents with Microsoft Copilot Studio to automate the generation of structured technical protocols of approximately 30 pages, subject to content and formatting constraints.
- Data analysis: analyzed healthcare reimbursement data to support strategic decisions, using LangChain, Excel and presentation tools.
- Used Git for source-code version control.

Capgemini Engineering — B-Reality *Sep. 2025 — Feb. 2026*
Research and Development Intern *Meudon, France*

- EEG signal classification for a c-VEP brain-computer interface integrated into a virtual reality rehabilitation system.
- Improved an existing eCCA method, evaluated neural classifiers and integrated models into a real-time processing pipeline.
- Experimental results: up to approximately 95% classification accuracy, with inference below 10 ms per cycle.
- Used Git for source-code version control.

Publication — Co-first author, IEEE ICCRAIDS 2026.
Optimization of cVEP Classification Approaches for 3D VR Based Brain Computer Interfaces.

Projects — Embedded Systems and Software

Processor Architecture and RISC-V Analysis
Academic team project · December 2024 · Git
– Developed a Python decoder for RISC-V instructions; studied a program adding two arrays, rewrote it in C and compared GNU toolchain output at -O0, -O and -O3.
– Analyzed a five-stage RISC-V pipeline: dependencies, stalls, forwarding and flushes. Designed a separate 16-bit instruction set in the report, with 16 registers and an IF/ID/EX three-stage architecture diagram.
– Work focused on instruction sets, assembly and architectural schematics.

STM32 Programming in C
Academic project · low-level programming · Git
– Brought up an initially unconfigured STM32 board: memory addressing, hardware initialization, GPIO and UART configuration, interrupt handling and peripheral control.
– Hands-on bare-metal C work connecting hardware registers, software and the observed behavior of the board.

STM32 Programming in Rust
Academic project · low-level programming · Git
– Developed embedded applications in Rust on STM32, including hardware configuration, interrupt handling and peripheral control.
– Explored a second language under the same hardware-access and embedded-execution constraints.

FPGA Memory Controller with an Avalon Interface
Academic project · SystemVerilog · Git
– Designed an Avalon-compatible memory controller and verified it on an FPGA board.
– Practiced RTL design, component communication and validation of a hardware interface.

Programmable Alarm Clock
Academic team project · Git
– Developed and verified an embedded programmable alarm clock through iterations in an Agile team workflow.
– Worked on system behavior, integration and functional verification.

Java Robot Simulator
Academic project · distributed systems · Git
– Built a robot simulator using a client-server architecture, middleware integration, and HTTP and socket communication.
– Applied object-oriented programming and integrated communication between the different parts of the application.

Raspberry Pi Remote-Controlled Car
Hardware and software project · Git
– Built a car controlled remotely through a web interface, using system programming on Raspberry Pi.
– Combined hardware and software integration, SSH access, Git version control and object-oriented design.

Projects — Artificial Intelligence and Data

Q-Learning Agent — Chartichaud
PROJ104 academic project · team of six
– Started with experiments on Pong, then developed a PettingZoo multi-agent environment and a Q-learning agent for the strategy game Chartichaud.
– Defined states, actions and rewards, and explored policy learning using Python, PyTorch and Gym. The poster documents the team project and its experiments.

OptiFrame — Lens Measurement and Frame Generation
CodeML hackathon · October 2026 · team project
– Web application that measures lenses from photographs and generates a parametric frame for 3D printing.
– Metric calibration with an ID-1 card and a homography; integration of pretrained YOLO-World v2 for detection and quantized SlimSAM for segmentation.
– Contour measurement, multiple-image processing, and STL or life-size SVG export. Processing runs locally in the browser after the models are downloaded: React, TypeScript, ONNX Runtime Web, WASM/WebGPU and Manifold3D.

Handwritten Digit Recognition
Personal project · NumPy, PyTorch, Pygame
– Explored a 784–32–32–10 NumPy network with sigmoid activations, handwritten derivatives and parameter updates; this implementation remains exploratory.
– Separately developed a PyTorch CNN for MNIST, using two convolutional layers, ReLU, max pooling, Adam and cross-entropy loss. A Pygame interface accepts a drawn digit and displays the three most likely classes.

Binary Classification of Face Images
Personal experiment · PyTorch
– Trained a convolutional network on 128 × 128 grayscale face images, using the two annotated classes in the dataset.
– Worked on image preprocessing and convolutional classifier training.

WhatsApp Conversation Analysis
Personal project · Python, natural language processing
– Built a tool for group-conversation statistics and positive, negative or neutral sentiment analysis using Hugging Face models.
– Practiced text processing, data organization and presentation of analysis results.

Technical Skills and Languages

AI and data: Python, NumPy, PyTorch, Pandas, scikit-learn, Hugging Face, SQL; reinforcement learning.
Systems and software: C and Rust on STM32, assembly, SystemVerilog/FPGA; Java, HTTP, sockets; React, TypeScript.
Tools: Linux, Git, Docker, Gurobi; Microsoft Copilot Studio.
Languages: French: native (C2). English: B2.2.

Association Responsibilities

Vice-President of Télécom Gaming Club, June 2024–June 2025. Coordinated a 20-member team, a tournament with over 100 participants, a budget of approximately €8,000 and partner relationships.